Decimals

Number · Edexcel IGCSE Mathematics A (4MA1) · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 62 minutes · Total marks: 62

KEY RULES

Ordering	Compare digit by digit from the left; add zeros so lengths match
Add / subtract	Line up the decimal points
Multiply	Ignore the points, multiply, then give the answer the same total decimal places
Divide	Multiply both numbers by 10, 100, ... until the divisor is a whole number
Fraction to decimal	Divide the numerator by the denominator: $\frac{3}{8} = 3 \div 8 = 0.375$
Recurring to fraction	$x = 0.\overline{45}$: $100x - x = 45$, so $x = \frac{45}{99} = \frac{5}{11}$

WORKED EXAMPLE 1

[3 marks]

Write $0.\overline{45}$ as a fraction in its simplest form.

Step 1	Let x equal the recurring decimal	$x = 0.454545\ldots$
Step 2	Multiply by 100 (two digits repeat)	$100x = 45.454545\ldots$
Step 3	Subtract to remove the repeating part	$99x = 45$
Step 4	Divide both sides by 99	$x = \frac{45}{99}$
Step 5	Simplify (divide by 9)	$x = \frac{5}{11}$

Answer: $\frac{5}{11}$

WORKED EXAMPLE 2

[2 marks]

Work out 3.6×0.25 without a calculator.

Step 1	Ignore the decimal points	$36 \times 25 = 900$
Step 2	Count decimal places: 1 + 2	3 decimal places
Step 3	Put the point back in	0.900

Answer: 0.9

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WORKED EXAMPLE 3

[2 marks]

Write these in order, smallest first: $0.7,\ \frac{5}{8},\ 0.65,\ \frac{2}{3}$

Step 1 Change each to a decimal — $0.7,\ 0.625,\ 0.65,\ 0.667\ldots$

Step 2 Compare digit by digit — $0.625 < 0.65 < 0.667 < 0.7$

Answer: $\frac{5}{8},\ 0.65,\ \frac{2}{3},\ 0.7$

1 Work out $4.7 + 2.35$

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

2 Work out $12.5 - 4.78$

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

3 Work out 0.8×0.6

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

4 Write $\frac{5}{8}$ as a decimal.

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

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5 Write 0.35 as a fraction in its simplest form.

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

6 Work out $3.2 \div 0.4$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

7 Work out 1.5×2.4

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

8 Work out 0.15×0.36 without a calculator.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

9 Work out $6.4 \div 0.16$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

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10 A bottle holds 1.25 litres. Glasses hold 0.05 litres each.

How many glasses can be filled from the bottle?

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

11 Write these in order, smallest first:

$0.45, \quad \frac{3}{8}, \quad 0.405, \quad \frac{7}{16}$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

12 Write $0.\dot{7}$ as a fraction.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

13 Write $0.\dot{1}\dot{8}$ as a fraction in its simplest form.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

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14 Pens cost £1.35 each. Sam buys 12 pens.

He pays with a £20 note.

Work out how much change he should get.

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

15 Use algebra to show that $0.7\dot{6}\dot{3} = \frac{42}{55}$

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16 Use algebra to show that the recurring decimal $0.2\dot{5}\dot{4} = \frac{14}{55}$

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17 Use algebra to show that $0.3\dot{7}\dot{8} = \frac{25}{66}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2025 · Question 16

18 Use algebra to show that the recurring decimal $0.6\dot{1}\dot{2} = \frac{101}{165}$

(Total for question = 2 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2025 · Question 16

19 Use algebra to show that $0.1\dot{7}\dot{6} = \frac{35}{198}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2023 · Question 16

20 Use algebra to show that $0.5\dot{3}\dot{9} = \frac{89}{165}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2026 · Question 17

21 Use algebra to show that $0.3\dot{4}\dot{5} = \frac{19}{55}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2022 · Question 17

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22 Show that the recurring decimal $0.1\dot{7} = \frac{8}{45}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2012 · Question 17

23 Use algebra to show that $0.\dot{7}0\dot{2} = \frac{26}{37}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 18

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24 Use algebra to show that $0.\dot{3}0\dot{6} = \frac{34}{111}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 18

25 Show that the recurring decimal $0.\dot{3}9\dot{6} = \frac{44}{111}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2011 · Question 18

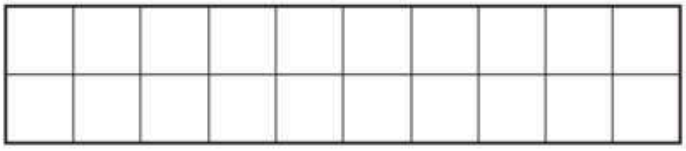
26 Show that the recurring decimal $0.0\dot{1}\dot{5} = \frac{1}{66}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2014 · Question 20

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CHALLENGE QUESTIONS

27 Here is a rectangle made from squares.



(a) Shade 0.7 of the rectangle.

(1)

(b) Write down the value of the 2 in the number 3.289

.....

(1)

(c) Write $\frac{5}{8}$ as a decimal.

.....

(1)

(d) Write these numbers in order of size.

Start with the smallest number.

2.803 2.008 2.081 2.83 2.8

.....

(2)

(Total for question = 5 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2022 · Question 3

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28 Use algebra to show that $0.68\dot{1}\dot{1} = \frac{15}{22}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2020 · Question 13

29 Use algebra to show that the recurring decimal $0.28\dot{1}\dot{3} = \frac{557}{1980}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2021 · Question 16

30 Use algebra to show that $0.4\dot{3}\dot{8} = \frac{217}{495}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 16

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31 $0.4\dot{x}$ is a recurring decimal.
x is a whole number such that $1 \leqslant x \leqslant 9$

Find, in terms of x, the recurring decimal $0.4\dot{x}$ as a fraction.
Give your fraction in its simplest form.
Show clear algebraic working.

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2021 · Question 18

TOTAL FOR PAPER IS 62 MARKS

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